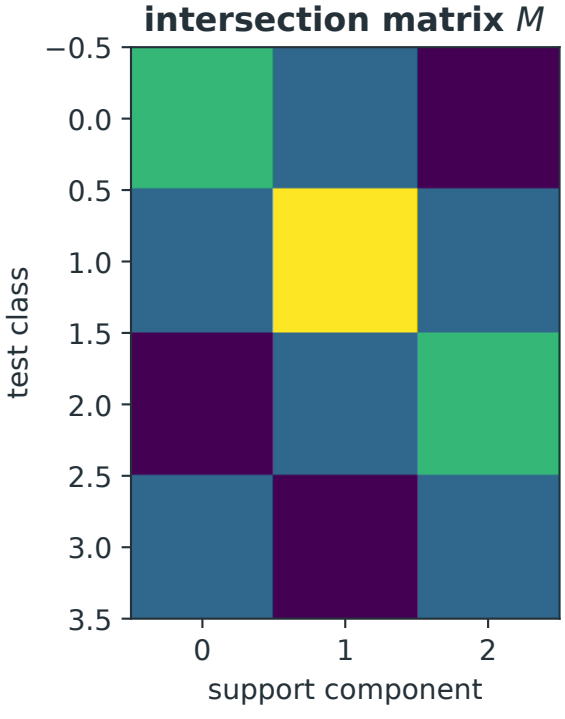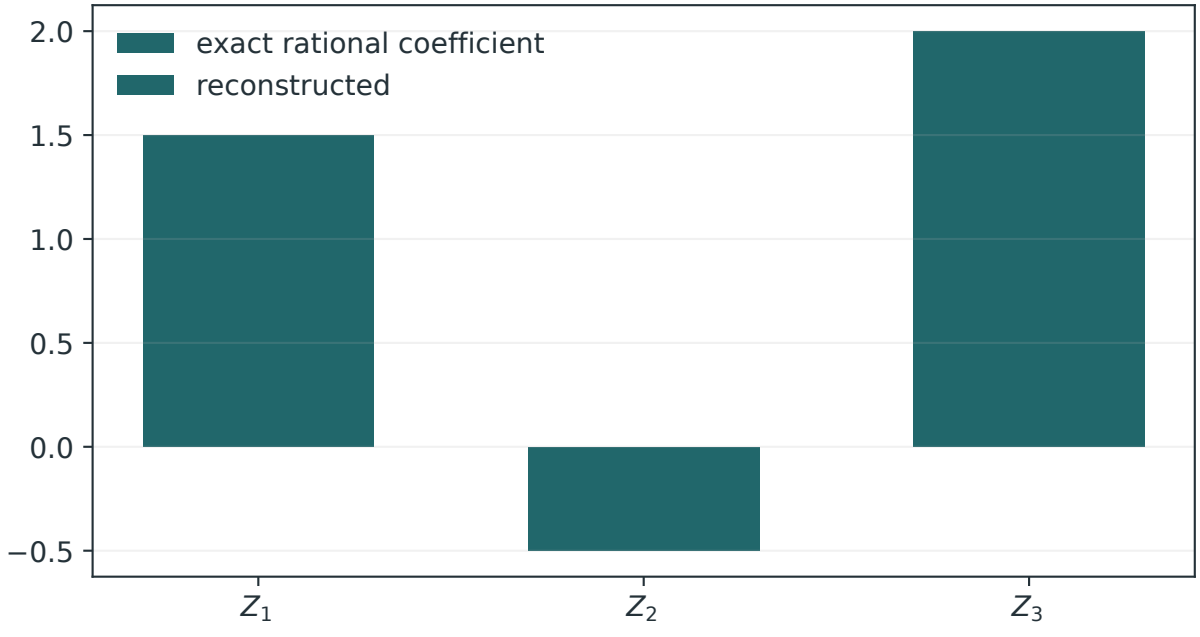Once the support is known, multiplicities are finite rational linear algebra